Information Resources Management Journal
Volume 38 • Issue 1 • January-December 2025

ChatGPT in the Working World:
A Qualitative Study

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ABSTRACT

The authors investigated how ChatGPT transforms workplace tasks by analyzing qualitative survey responses from 78 U.S. professionals in the fields of software, marketing, and academia. This study addresses a gap in understanding the malleability of generative artificial intelligence in diverse professional contexts, a need underscored by ChatGPT's rapid adoption and mixed impact on work practices. Using a qualitative survey deployed via a validated platform, the authors collected open-ended responses about tasks, challenges, and opportunities. Responses were inductively coded to compare domain-specific applications. The findings show that although ChatGPT is widely used across sectors for tasks such as content creation, research, and idea generation, certain tasks—such as coding in software, strategic communication in marketing, and knowledge acquisition in academia—diverge. The results emphasize the importance of context-sensitive integration strategies and bottom-up adoption approaches to maximize the benefits of artificial intelligence while mitigating risks.

KEYWORDS

Generative AI, Technology Adoption, Human-AI Interaction, Organizational Malleability, Qualitative Research, ChatGPT, Job Displacement, Future Work

INTRODUCTION

Generative artificial intelligence (AI), especially ChatGPT, has rapidly permeated the world, leading to an unprecedented wave of global exploration and trial and error in technology adoption that is driven by its accessibility and ease of use across diverse user groups. Surveys have shown that in 2024 approximately 62% of employees used ChatGPT or similar tools for work-related tasks, an increase from 27% in November 2022 and 43% in January 2023 (Glassdoor Economic Research, 2024). In contrast to purpose-specific software solutions designed for specific functions, such as customer relationship management, enterprise resource planning, and business intelligence—and typically requiring tailored training for select employees—current large language models (LLMs)

DOI: 10.4018/IRMJ.386593

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represent a new class of general-purpose technology. These models are characterized by inherent flexibility and openness, which enable users to complete an extensive range of tasks that encompass almost any activity that can be articulated in natural language (Bilgram & Laarmann, 2023; Felten et al., 2023). The performance of OpenAI's latest language model on numerous standardized tests (e.g., the GRE, SAT, and Uniform Bar Examination) ranks in the 90th percentile, reflecting its breadth of capabilities and competitiveness (OpenAI, 2023). This broad applicability and ease of access introduce both opportunities and complexities given that organizations are still in the nascent stages of fully integrating these tools into daily workflows.

In information systems research, the extent to which a system can be shaped, modified, or adapted to meet different user needs, contexts, and applications is referred to as *malleability* (Richter & Riemer, 2013). Despite this widespread deployment and significant impact, current research has provided limited insights into the specific tasks LLMs are transforming across professional domains. Equally underexplored are employees' opportunities and challenges as they adopt LLMs in their roles, highlighting a gap in understanding how these models reshape the modern workplace. Moreover, ignoring the malleability of LLMs is a critical gap in current research because opportunities and challenges related to the use of LLMs may be contingent on the specific task for which they are used (Goodhue & Thompson, 1995). A deeper understanding of these interdependencies is critical because organizations without a comprehensive view of the technology's application fields risk inefficiencies, missed opportunities for innovation, or poor strategy development. Furthermore, a lack of awareness around the specific challenges and opportunities that ChatGPT presents could lead to inadequate responses, such as insufficient training or poorly designed employee education programs.

Although some organizations have issued general guidelines for using ChatGPT, these often emphasize mitigating risks, such as protecting data privacy and safeguarding confidential information, rather than focusing on harnessing the technology's full potential. In practice, employees are already leveraging generative AI tools in various ways, whether through official company channels or personal accounts used to boost productivity as a "secret weapon." A notable 70% of employees who use ChatGPT report using it without prior alignment with their supervisors (Jackson, 2023). This widespread and partly unofficial use underscores the pressing need for a more nuanced understanding of and strategic approaches to ChatGPT's role in the workplace.

To fill these research gaps, this study addressed two key questions. First, which tasks are being transformed by ChatGPT in different professional fields? Second, what challenges and opportunities does the use of ChatGPT create for these sectors? Drawing on qualitative survey data from 78 employees across three professional fields identified as high risk for AI impact on the basis of the AI Occupational Exposure (AIOE) index (Felten et al., 2023), we examined how employees integrate this emerging technology; specifically, we identified tasks, opportunities, and challenges related to using ChatGPT. Employees participating in this study used ChatGPT-3.5 (65%), ChatGPT-4.0 (22%), both (9%), or did not know the model version (4%).

This study contributes to a better understanding of the malleability of ChatGPT from an employee perspective across different sectors. Our findings offer a comprehensive overview of how ChatGPT is being integrated into the modern workplace and outline the various tasks it supports in marketing, software development, and academia. Furthermore, we uncover important challenges related to the technology, such as concerns around accuracy, overreliance, and job security, which vary across sectors.

LITERATURE REVIEW

AI refers to systems designed to perform tasks that traditionally have required human intelligence, such as problem solving, decision making, and learning by mimicking human cognitive abilities (Fügener et al., 2021; Russell et al., 2010). A prominent subset of AI is *machine learning*, whereby systems improve over time by learning from data (Sebastiani, 2002). Within machine learning, natural language processing enables machines to understand and generate human language,

allowing for an easy way of interacting (Liu et al., 2017). Transformer architecture has recently led to the introduction of powerful LLMs by enabling AI models, such as GPT-4, to process and generate language by using self-attention mechanisms to weigh the importance of different words in a sequence, allowing for efficient parallel processing and deep contextual understanding (Brown et al., 2020; Vaswani et al., 2017). Chatbots like ChatGPT can generate humanlike text, images, videos, and code, making it useful for a variety of tasks, such as content creation, research, creative work, and communication.

Although AI models have traditionally been designed for specific use cases, often requiring extensive training and customization for each unique application (Kakatkar et al., 2020; Marchuk et al., 2020), foundation models represent a shift toward context-independent, universal AI with robust pretrained models that are capable of undertaking a broad array of tasks without additional specialized training (Bommasani et al., 2021). This shift reduces the time and resources that organizations need to invest in order to seamlessly apply AI across multiple functions. Its accessibility and ease of use have contributed to its widespread adoption in workplaces, in particular for tasks that require the efficient generation or processing of large amounts of text (Teubner et al., 2023). The combined effect of democratization and the universal suitability for a wide variety of tasks have significantly expanded ChatGPT's reach and impact within organizations (Biel et al., 2025).

The task–technology fit framework posits that a technology's effectiveness depends on the degree to which its functionalities align with the demands of a specific task (Goodhue & Thompson, 1995). This model traditionally has assumed a relatively stable relationship between a technology's features and its optimal use case, suggesting that systems are most effective when tailored to narrowly defined tasks (Zigurs & Buckland, 1998). However, recent advances in generative AI, such as ChatGPT, challenge these assumptions. As a highly malleable system, ChatGPT enables users to shape its outputs through prompt design and contextual knowledge, allowing it to serve both cross-functional and domain-specific purposes. This interpretative flexibility shifts the locus of fit from the technology itself to the interaction among user intent, domain expertise, and adaptive use, suggesting a need to extend task–technology fit theory to better account for user-driven appropriation and tool plasticity (Strong et al., 2006).

LLMs, such as OpenAI's GPT-4 model, on which ChatGPT is based, are reshaping diverse professional fields by enabling new levels of task automation and efficiency as well as creativity (Bilgram & Laarmann, 2023; Chen et al., 2023; Polster et al., 2024). Thus far, researchers who have examined the impact of generative AI on the field of work have mostly focused their investigations on a specific task or profession (Boussioux et al., 2024; Champa et al., 2024; Kshetri et al., 2024). A key area of scholarly interest is the comparison of human performance and AI, which spans numerous disciplines and has demonstrated impressive results. For example, GPT-4 successfully passed the Uniform Bar Examination, a standardized test that measures the knowledge and skills required for legal practice in the United States (Katz et al., 2024). In another instance, ChatGPT outperformed human experts in generating ideas for a specific briefing, excelling in both novelty and customer value (Joosten et al., 2024). Similarly, consultants at the Boston Consulting Group achieved outcomes that were over 40% improved and completed their work 25% faster with ChatGPT support (Dell'Acqua et al., 2023). Furthermore, a recent study showed that the demand for freelancers in automation-prone roles, such as writing and coding, dropped by 21% within 8 months of ChatGPT's release, indicating that certain jobs may be increasingly at risk of substitution by AI (Demirci et al., 2024).

Other studies have extended their focus from one specific task (e.g., idea generation or copy creation) to a broader professional field (e.g., marketing or software development). In software development, for example, ChatGPT serves as a valuable tool for code generation, debugging, feature development, and problem solving (Champa et al., 2024; White et al., 2024; Xiao et al., 2024). Developers use ChatGPT to automate repetitive tasks and accelerate the coding process, allowing them to focus on adding value to software projects (Ebert & Louridas, 2023; Fraiwan & Khasawneh, 2023). An empirical study conducted by Tian et al. (2023) suggested ChatGPT's usefulness for code

generation, program repair, and code summarization within software development based on a comparison with benchmark approaches. Similarly, Sobania et al. (2023) found that automatic bug fixing by ChatGPT achieved performances similar to those of several dedicated automated program repair approaches.

Despite the usefulness of ChatGPT, however, limitations exist, such as inaccuracies in generated code, manual error correction, and cybersecurity risks from potentially malicious code (Abdullah et al., 2022). As AI technology advances, its role in reshaping software development is expected to grow, although the short-term impact on programming careers remains uncertain (Rahmaniar, 2024).

Research on the use of AI in marketing professions highlights cases in content creation, lead generation, personalization, customer service, and customer engagement (Fraiwan & Khasawneh, 2023; Gupta et al., 2024; Kshetri et al., 2024). AI's ability to generate personalized marketing messages and analyze consumer behavior allows businesses to make data-driven decisions, optimize customer engagement, and reduce operational costs (Haleem et al., 2023; Haluza & Jungwirth, 2023; Taecharungroj, 2023). Two experiments conducted by Zhang and Prebensen (2024), for instance, have demonstrated that AI-generated texts result in tourist evaluations that are similar to human-generated content. However, privacy concerns, misinformation, and bias remain critical challenges, especially when ChatGPT is used to process sensitive consumer data or when it generates inappropriate responses (Eliot, 2023; Johnson, 2022; Paul et al., 2023).

ChatGPT has also drawn attention from academics across various fields. It assists teachers by automating tasks such as grading, generating feedback, and creating instructional materials (Bahrini et al., 2023; George & George, 2023). In academic research, ChatGPT is mainly used to assist in writing, reviewing, and synthesizing related work; identifying research questions; sampling; collecting data; and interpreting those data (Gregor, 2024; Ngwenyama & Rowe, 2024). For example, ChatGPT could assist in the development of research questionnaires (Abdullah et al., 2022) or literature reviews (Ngwenyama & Rowe, 2024).

Although AI-driven tools such as ChatGPT offer promising possibilities for accelerating research and assisting with academic tasks, they also raise ethical and practical concerns in this area. Overreliance on AI could lead to a proliferation of superficial research that lacks the depth and innovative thinking that truly advance a field (Lund et al., 2023). In terms of scientific integrity, the degree to which the integration of ChatGPT into research will lead to plagiarism is not yet clear (Dwivedi et al., 2023).

Although insightful, previous studies have several limitations we sought to address in this research. First, as mentioned, most articles focus on one specific context, such as a concrete task like idea generation (Boussioux et al., 2024; Joosten et al., 2024); a professional field, such as software development (Champa et al., 2024; Rahmaniar, 2024); or the general workforce without accounting for sector-specific dynamics (e.g., Retkowsky et al., 2024), thus inhibiting a holistic perspective. Although such research settings allow for deep insights, they fall short of comparing AI use across occupational or task-related boundaries. Second, our literature review revealed that many existing research papers are conceptual or theoretical and lack a broad empirical foundation to offer generalizable results and uncover patterns (Cotton et al., 2024; George et al., 2023; Gregor, 2024; Gupta et al., 2024; Kshetri et al., 2024; White et al., 2024). Third, previous studies have focused on different objects of investigation, such as diffusion of AI technologies (Choi et al., 2023; Ge & Wu, 2023; Gupta et al., 2023), human–machine interaction via prompting (Champa et al., 2024; Xiao et al., 2024), and the consequences of AI use on perceived productivity and sense of accomplishment (Kobiella et al., 2024; Miyazaki et al., 2024).

Few studies have thoroughly examined the specific tasks or domains in which AI is used. A notable exception is Giordano et al.'s (2024) quantitative analysis of Twitter data, which explored the tasks and skills influenced by ChatGPT. They identified 408 distinct tasks and mapped them to 185 skills potentially impacted by AI. Moreover, their findings illuminate user perceptions of ChatGPT's performance, with a sentiment analysis revealing that tasks such as enhancing customer interactions,

providing constructive feedback, and brainstorming ideas were among the most positively regarded capabilities. Though insightful, Giordano et al.'s study suffers from a data source bias because it relied on Twitter as a primary source of data. Because tweets often include casual, leisure-oriented, and humorous content (e.g., writing songs or jokes and creating digital images), their study revealed the broader use of ChatGPT in society rather than focusing on its application in professional settings.

To address these gaps, in this research we adopted a qualitative survey-based methodology to provide a more systematic and targeted understanding of ChatGPT's applications across diverse professional sectors. By collecting detailed, context-specific responses from professionals, we explored work-related use cases and their implications. We then inductively coded and analyzed these responses to identify recurring patterns and emerging themes regarding the roles ChatGPT plays in supporting human skills. This approach also allowed us to systematically compare the use of ChatGPT among individuals in different professional domains.

METHODOLOGY

To define the professional fields for our survey, we drew on the AIOE index, which estimates the extent to which occupations are likely to be affected or transformed by AI (Felten et al., 2021, 2023). The index is based on an expert analysis that matches AI capabilities with job requirements. Our study focused on occupations susceptible to ChatGPT's influence, specifically those with medium to high AIOE scores. We therefore chose three fields, in line with Chiong and Xie's (2024) study, which defined medium to high AI exposure groups to examine AI's impact on job satisfaction. The three fields are categorized as follows: (a) software, with AIOE scores $\geq$0.70; (b) marketing, with AIOE scores of $\geq$0.80; and (c) academia, with AIOE scores $\geq$0.90.

We used Prolific, a validated and well-established survey platform (Palan & Schitter, 2018), to prescreen and distribute the survey. Douglas et al. (2023) found that Prolific, alongside CloudResearch, provides the highest data quality compared with competitors such as MTurk, Qualtrics, and SONA. A purposive sampling strategy was applied using Prolific's sample creation tools. Participants were required to be based in the United States, employed full or part time, and be regular users of screen-based devices. The sample was limited to the United States because it is the country with the highest number of ChatGPT users (Duarte, 2023). In addition, U.S. data protection regulations are less restrictive than those in the European Union (Pop, 2023; Schwartz, 2019), which may increase the likelihood of ChatGPT being used in professional contexts.

A prescreening survey was embedded via Prolific and used to filter the sample by asking whether ChatGPT is occasionally used for work and in which sector the respondent is employed. The prescreening survey was completed by 500 individuals. Of those, 41 participants identified with academia, 58 with marketing, and 71 with software development. From this pool, 30 professionals per sector were invited to participate in the main survey to ensure a minimum of 20 valid responses per sector (cf. Guest et al., 2006) and achieve theoretical saturation, understood as the point at which no new insights emerge from additional data (Guest et al., 2006).

The main survey, conducted in September 2023 and created with SoSci Survey (Version 3.4.22; Leiner, 2023), consisted of four background questions (i.e., that inquired about ChatGPT version, usage frequency, language of interaction, and job title). Usage frequency was measured with a 7-point Likert scale (1 = *once per working month*, 7 = *hourly per working day*) because this optimizes reliability, validity, and response differentiation (Taherdoost, 2019). Although all participants were based in the United States, we included language of interaction as a demographic variable to account for the possibility that multilingual professionals may interact with ChatGPT in a language other than English (e.g., their native language). This ensured that responses could be interpreted in light of any potential linguistic variation in the interactions. Furthermore, to answer the research questions, the survey consisted of three qualitative open-ended questions that addressed three categories (i.e.,

tasks, challenges, opportunities) that were deductively determined on the basis of related research (Dwivedi et al., 2023; Haleem et al., 2023; Ray, 2023).

To enable a robust exploration of how ChatGPT is embedded in daily professional tasks, as well as the challenges and opportunities it may present, we encouraged participants to provide detailed answers. As an incentive, clearly communicated within the questionnaire alongside the open-ended questions, participants were offered a 25% bonus payment for meeting the following word thresholds: at least 45 words for task elaboration and at least 75 words each for describing challenges and opportunities. This incentive was designed to encourage more elaborate responses by emphasizing the value of detailed input. According to available studies, offering incentives is unlikely to change or negatively affect the quality of a survey (Singer & Ye, 2013).

To further ensure data quality, we applied strict exclusion criteria: Questionnaires with syntactical overlap (e.g., very similar sentences) in the responses were excluded (in this case, two questionnaires). Furthermore, if questions were insufficiently, irrelevantly, or otherwise incorrectly answered, those questionnaires were also removed (in this case, six questionnaires). Responses were verified for similarities with existing content on the web using the Turnitin plagiarism detection service via Scribbr (2024). This check did not involve AI authorship detection but focused only on identifying copied or nonoriginal content because AI authorship cannot reliably be detected (Sadasivan et al., 2023). In cases of confirmed plagiarism, the questionnaire was excluded (in this case, four questionnaires). As mentioned, the questionnaire was organized around three deductively defined blocks: tasks, challenges, and opportunities. Building on that structure, we applied the following two-step categorization procedure during analysis. In Step 1 (deductive frame), the three blocks provided the initial top-level categories. In Step 2 (inductive code generation), we conducted inductive content analysis within each block following Mayring's (2014) guidelines, using MAXQDA Analytics Pro 2022 (Release 22.8.0; MAXQDA, 2023) to manage and document the coding process. Thus, all specific labels (e.g., content and copy creation, accuracy and reliability issues) directly arose from the data and were not imported from prior taxonomies. AI-assisted features of the software were not used; all codes were manually generated through human interpretation.

To ensure reliability of the qualitative coding, all 510 coded segments were cross-validated by a second person with subject matter expertise (i.e., a researcher knowledgeable in the generative AI field). To avoid double counting, any given code could be assigned at most once per answer, although multiple codes could be applied to a single response. For further details regarding the coding process, see Figure A1 in the Appendix. Furthermore, Appendix Tables A1–A3 include explanations for each code, and illustrative examples are also provided for tasks. These clarify how inductively derived categories (e.g., research and data analysis) were defined and applied. Finally, once all the data were coded, we compiled them into concept matrices (Webster & Watson, 2002), enabling a clear visualization of the frequency of each code within the three sectors.

RESULTS

Background information on participants from the sectors of academia, marketing, and software development are presented in Tables 1–3, and the identified tasks, challenges, and opportunities, are provided in Tables 4, 5, and 6, respectively.

Descriptive Statistics

The demographic profile of the surveyed professionals is provided in Tables 1 and 2.

Table 1. Participants' Demographic Characteristics

Characteristic	A	MA	S
Total	29	25	24
Gender			
Male	55.17	68.00	83.30
Female	44.83	32.00	16.70
Student status	31.03	12.00	16.70
Employment status			
Full time	86.21	84.00	95.80
Part time	13.79	16.00	4.20

Note. All table values are percentages. A = academia; MA = marketing; S = software development.

The male:female ratio and student/employment statuses were inferred on the basis of data provided by Prolific. Further details on the age distribution of the participants are provided in Table 2.

Table 2. Age Distribution Across Sectors

Sector	*M*	*SD*	Min	*Mdn*	Max
A ($n = 29$)	38.21	9.02	23	37	60
MA ($n = 25$)	39.60	12.18	19	37	62
S ($n = 24$)	36.58	8.55	23	35.5	61

Note. Min = minimum; Max = maximum; A = academia; MA = marketing; S = software development.

Age distribution was also inferred on the basis of information provided by Prolific. To further contextualize respondents' backgrounds, we also included additional background questions in the survey. The results are presented in Table 3.

Table 3. Background Information on Participants' Use of ChatGPT

ChatGPT usage characteristic	A ($n = 29$)	MA ($n = 25$)	S ($n = 24$)
Usage frequency (*M*/*SD*)	3.76/1.75	4.20/1.55	4.75/1.26
Use of English language	29	25	24
ChatGPT-3.5	21	17	13
ChatGPT-4.0	6	4	7
ChatGPT, both versions	0	3	4
Do not know the ChatGPT version	2	1	0

Note. A = academia; MA = marketing; S = software development.

The results in Table 3 show a mean frequency of using ChatGPT of 3.76, with a standard deviation of 1.75 in academia. For marketing, the average is 4.20 ($SD = 1.55$), and software development has the highest frequency of use with a mean value of 4.75 ($SD = 1.26$). When asked about the version of ChatGPT used, 21 participants in academia reported using ChatGPT-3.5, and six reported using Version 4.0. In marketing, 17 participants used ChatGPT-3.5, four used Version 4.0, and three reported using both versions. In software development, 13 participants were familiar with ChatGPT-3.5, seven were familiar with Version 4.0, and four participants who used both versions. In addition, some participants were unsure which version they were using: two in academia, one in marketing, and none in software development. Participants were free to indicate the language in which they interacted with ChatGPT; all reported primarily using English.

Tasks Identified

Eight tasks were inductively identified, with respondents mentioning an average of 2.03 tasks. Table 4 is structured in accordance with total mentions, highlighting the most frequently reported activities.

Table 4. Tasks Performed Using ChatGPT

Code	A	MA	S	Total
Content and Copy Creation	55.17	80.00	20.83	52.56
Research and Data Analysis	31.03	52.00	16.67	33.33
Idea Generation	31.03	28.00	37.50	32.05
Software Development and Coding	17.24	0.00	58.33	24.36
Knowledge Acquisition	34.48	12.00	20.83	23.08
Stakeholder Communication	17.24	20.00	12.50	16.67
Administrative and Operational Support	17.24	12.00	20.83	16.67
SEO and Keyword Research	0.00	12.00	0.00	3.85

Note. All table values are percentages. A = academia; MA = marketing; S = software development; SEO = search engine optimization.

Content and Copy Creation emerged as the most frequently mentioned task among all those identified. In academia, this was commonly described as creating tailored learning materials. For example, one participant stated, "I use ChatGPT to create engaging math problems that I can give my students to work on." Moreover, participants reported using ChatGPT to revise academic papers for clarity or to create professional text, such as job descriptions. In marketing, participants used ChatGPT for creating blogs, social media, and advertising content. In software development, tasks such as ensuring text accuracy, drafting documents, and generating social media posts are also included under this code.

Research and Data Analysis was the second most frequently mentioned task. This involves using ChatGPT to review literature, synthesize information, validate facts, analyze user feedback, cross-check methodologies, and identify trends or insights in large datasets. In academia, for example, a postdoctoral associate summarized how ChatGPT aids in "fact-checking, citation assistance, efficient literature review, peer review preparation." Marketing and software participants described using ChatGPT to extract information from customer feedback and to generate trend predictions on the basis of internal data.

Idea Generation was the third most frequently cited code. Participants reported using ChatGPT to brainstorm and generate ideas across various contexts. In academia, it can be used to generate

ideas for teaching methods or in academic writing. A Spanish lecturer, for example, used ChatGPT to design a treasure hunt game for students who were practicing indefinite Spanish expressions. In marketing, content creators said they used ChatGPT to brainstorm ideas and generate outlines for emails, landing pages, blog posts, and similar materials. In software development, participants reported using ChatGPT to develop technical ideas, as illustrated by a participant whose team consulted ChatGPT to understand and solve a video player issue after an update. One data architect noted, "It does provide a good catalyst to help me move toward where I want to arrive."

Software Development and Coding was reported almost exclusively by participants in the software sector. Respondents described using ChatGPT to develop algorithms, write or optimize code snippets, troubleshoot software problems, and refine existing solutions. Specific uses included explaining code blocks, suggesting edits, and rewriting code for improved clarity or functionality. Some participants also reported using ChatGPT to generate project documentation and recreate user-interaction scenarios for usability checks.

Knowledge Acquisition refers to the use of ChatGPT to acquire knowledge and retrieve information. Participants said they used ChatGPT to understand concepts, summarize literature, answer questions, and support other educational tasks. Professionals across all three sectors used it to quickly access information. Academics reported using it to summarize literature, suggest additional sources, and simplify complex texts; lecturers also used it to guide students who encountered difficulties. Marketing professionals applied ChatGPT to obtain definitions and information on current industry trends. Software developers used it to clarify functions, concepts, and unfamiliar terms.

Additional Tasks

Given their lower prevalence (<20% of respondents), these tasks are concisely summarized to maintain focus on the most prominent tasks.

Stakeholder Communication included drafting personalized emails; providing real-time chat support; responding to inquiries; and polishing written correspondence for spelling, grammar, and style.

Administrative and Operational Support involved report writing, data summarization, information organization, task automation, project planning, performance management, and the facilitation of employee and student evaluations.

SEO (Search Engine Optimization) and Keyword Research tasks mentioned by respondents included generating keyword suggestions, optimizing search engine content, and providing insights for blog posts or advertising campaigns to enhance search rankings.

Identified Challenges

Participants' challenges in using ChatGPT were identified as follows.

Table 5. Challenges in ChatGPT Usage

Code	A	MA	S	Total
Accuracy and Reliability Issues	37.93	44.00	45.83	42.31
Operational and Implementation Challenges	34.48	36.00	54.17	41.03
Overreliance and Deskilling	44.83	20.00	37.50	34.62
Originality and Plagiarism Concerns	41.38	28.00	16.67	29.49
Job Replacement Anxiety	17.24	28.00	41.67	28.21
Privacy and Data Security Concerns	20.69	20.00	37.50	25.64

continued on following page

Table 5. Continued

Code	A	MA	S	Total
Social and Economic Impact	10.34	8.00	8.33	8.97
AI Bias and Misuse	3.45	8.00	12.50	7.69

Note. All table values are percentages. A = academia; MA = marketing; S = software development; AI = artificial intelligence.

Accuracy and Reliability Issues were commonly reported as concerns with respect to the trustworthiness of ChatGPT's output. These issues were relevant across all sectors, yet their implications varied. Academics and educators noted that inaccuracies could compromise scholarly integrity, for example, by introducing calculation errors, mis-citing sources, or distorting learning materials. Respondents in the marketing sector expressed concerns that outdated or inaccurate information could mislead campaigns, damage brand reputation, and lead to a loss of customers. Last, software developers reported that erroneous code suggestions might introduce bugs, security vulnerabilities, or even cause system failures.

Operational and Implementation Challenges included difficulties professionals faced in crafting prompts that yield satisfactory outputs from ChatGPT. The tool's generic tone may also hinder personalization for individual users, which could lead to impersonal responses unless users adapt the output. In addition, some professionals reported that reluctance among certain employees to use AI may impede its full utilization. Because ChatGPT's output does not always guarantee factual accuracy, many professionals said they dedicated additional time to checking the results before final use. Moreover, the professionals stated that ChatGPT is a developing tool that requires new usage knowledge and continuous learning to maximize its potential. Participants noted that in professions dealing with potentially discriminatory topics (e.g., journalism), sensitive issues, or data that require a high level of sensitivity (e.g., certain chemical compositions), ChatGPT often preventatively does not respond, leading to censorship barriers. Furthermore, ChatGPT occasionally may misinterpret context or hit data type and size limitations, which can hinder its effectiveness.

Overreliance and Deskilling were described by professionals as comprising concern about a decline in critical thinking and research skills due to excessive reliance on ChatGPT. They also noted that using it for assignments, creative writing, or coding may create fundamental knowledge gaps and undermine users' ability to learn, create, and innovate independently. Marketing respondents felt that overdependence on ChatGPT for customer service may reduce personal contact and negatively affect customer experiences.

Originality and Plagiarism Concerns were reported in all three sectors, although the specific implications differed. Academic respondents noted that ChatGPT could undermine original writing and research; educators worried that students might use the tool for assignments in ways that are difficult to detect. One student said they had been suspected of plagiarism because ChatGPT involvement was assumed, even though it had served only as inspiration. Academics were uncertain about where legitimate assistance ends and plagiarism begins. Marketing professionals emphasized that brand identity depends on distinctive, authentic copy; they feared ChatGPT's often generic output could homogenize content at scale and erode customer trust. Software sector participants questioned intellectual property ownership when using ChatGPT-generated code or text, noting that output might not be fully owned by the organization.

Job Replacement Anxiety was reported as a concern across all sectors, reflecting worries that ChatGPT might automate tasks usually carried out by humans, potentially resulting in job displacement, salary reductions, or a decrease in the value of certain roles . In academia, educators expressed concerns that ChatGPT could replace them or reduce salaries because it might take over certain instructional tasks and diminish the perceived value of teaching. Researchers shared these

concerns, fearing that improvements in ChatGPT's research capabilities might challenge their expertise. In marketing, participants noted that both creative and manual roles might be at risk, with some suggesting that lower level positions could be eliminated. Software developers mainly expressed anxiety about the displacement of technical skills, in particular for less experienced coders, as ChatGPT code generation capability improves. Some respondents also noted that job functions in customer service, copy generation, and accounting could face reduced demand as the tool becomes more efficient.

Privacy and Data Security Concerns were reported by respondents who feared that data could be compromised or regulatory requirements, such as General Data Protection Regulation, could be breached. Academics said that their intellectual property could be exposed because ChatGPT also relies on user input for model training. Marketing and software development respondents noted the risk of mishandling customer data, which could lead to breaches of the law, as well as the potential for data leakage when employees cut and paste sensitive company data.

Additional Challenges

Social and Economic Impact was reported by respondents, including the potential devaluation of academic roles and reduced peer interaction among students. Beyond academia, respondents noted that ChatGPT could influence hiring practices, work processes, and creative industries. Some participants raised concerns about compliance issues, in particular around General Data Protection Regulation and other regulatory frameworks. In addition, participants highlighted growing resource consumption, such as increased water use, which may rise with larger models.

AI Bias and Misuse were also challenges mentioned by respondents. One participant noted the difficulty of "mitigating algorithmic bias," and a marketing analyst emphasized the importance of preventing harmful or discriminatory content, ensuring that AI-generated messages align with an organization's values. Another respondent expressed the concern that AI could be exploited to manipulate opinions and behaviors, for example, by generating fake news articles or propaganda.

Identified Opportunities

The participants identified the following opportunities of using ChatGPT.

Table 6. Opportunities in ChatGPT Usage

Code	A	MA	S	Total
Efficiency and Timesaving	65.52	64.00	83.33	70.51
Content Proficiency and Data Intelligence	41.38	40.00	37.50	39.74
Learning Enhancement and Information Availability	55.17	8.00	37.50	34.62
Innovative Thinking and Creativity	44.83	24.00	29.17	33.33
Professional Development and Job Creation	13.79	32.00	29.17	24.36
Cost Efficiency and Profit Maximization	6.90	28.00	16.67	16.67
Global Communication, Inclusivity, and Collaboration Enhancement	10.34	16.00	16.67	14.10

Note. All table values are percentages. A = academia; MA = marketing; S = software development.

Efficiency and Timesaving comprised the most frequently cited opportunity. Respondents across sectors stated that ChatGPT automates routine tasks and streamlines workflows, leading to faster and more precise results, thereby freeing time for more strategic and creative work. Professionals

stated that "The clear advantage for myself using ChatGPT is time saving" and that "ChatGPT makes everything more efficient." Educators reported that automation of grading, administrative work, and lesson planning gave them more time to focus on lecturing; one lecturer remarked, "It can save time by adjusting the complexity of reading passages for students and crafting emails and newsletters for teachers." Researchers described using ChatGPT for content generation, grammar checking, and literature review support, which left more time for analysis. Marketing professionals reported benefits from streamlined content creation and rapid information retrieval. A project manager shared, "We don't have to spend tons of hours navigating traditional blogs or websites." In addition, they mentioned that ChatGPT automates data entry and analysis, which they believed could reduce errors and improve campaign efficiency. Respondents in the software sector mentioned quicker resolution of coding issues and automation of repetitive tasks; a director stated that ChatGPT helps the team "get through coding issues more quickly than we can do on our own."

With respect to Content Proficiency and Data Intelligence, ChatGPT was reported to enhance content creation and data analysis across sectors. Educators reported using it to expand curricula with real-world projects, aid grading, translate texts, and provide feedback (e.g., on tone and grammar), which some said improved engagement and comprehension. Marketing respondents noted that ChatGPT refines strategies, enhances personalization (e.g., A/B testing), and identifies trends in large datasets. It was also reported to assist with content creation (e.g., blog posts, social media copy). A senior digital media producer noted that AI-generated images based on prompts could aid in content production. Software development respondents reported that ChatGPT was used to generate code, reducing development time and improving quality. One participant highlighted benefits in supporting decision making by "finding important information in data and even suggesting better decisions." An operations engineer also noted its potential role in risk assessments, such as evaluating hardware replacements.

Learning Enhancement and Information Availability was an opportunity cited by participants. One respondent mentioned a form of self-study: "If used properly, students have a resource that can critique their work, offer advice, and help them understand complex tools." According to one lecturer's perspective, ChatGPT improves explanations and provides educational information. In addition, respondents highlighted that ChatGPT's on-demand information could benefit users: Instead of manually searching the web, it can directly provide answers tailored to the individual's queries. In marketing, rapid and specific information is vital, and ChatGPT's ability to retrieve tailored responses was noted as a valuable asset. In software development, professionals found ChatGPT's output easy to understand because it organizes content according to the query. Developers also mentioned leveraging ChatGPT for learning enhancement and skill-building.

With respect to Innovative Thinking and Creativity, academics and educators highlighted ChatGPT's ability to help generate new ideas and offer diverse perspectives, assisting them in critically reflecting on their own output, such as papers. They reported that students are encouraged to look beyond standard AI responses, thereby promoting critical thinking. ChatGPT was also seen as a tool for inspiring creativity in lecturing by challenging traditional learning methods. Marketers responded using it as a brainstorming device for creative tasks to explore new product ideas, refine marketing strategies, and address creative blocks. Software developers noted its support for creative problem solving. Furthermore, they mentioned that its analytical capabilities, such as identifying customer needs, could assist in generating new product ideas and fostering innovation.

Professional Development and Job Creation was stated as an opportunity resulting from ChatGPT. This comprised opening new career paths and enhancing existing roles. Academics reported developing coding, user experience, or data analysis skills; reducing time on routine tasks; and potentially increasing job satisfaction or career advancement, including promotions and pay raises. Marketers perceived career growth through prompt engineering, understood as a specialized skill that offers businesses a competitive edge, and through the creation of AI-focused departments. Developers

described ChatGPT as transformative, from building specialized software that integrates the AI tool to launching novel products, such as personalized learning tools or interactive marketing campaigns.

Additional Opportunities

Cost Efficiency and Profit Maximization were reported by participants in academia, where ChatGPT was said to accelerate publications by improving efficiency. Marketing professionals considered cost efficiency as reducing time spent on information searches, replacing salaried employees, optimizing inventory, or detecting fraud. Participants found that using ChatGPT is already contributing, or could contribute, to profit maximization. Software development participants noted increased sales and marketing through engaging content and cost savings by automating tasks.

Global Communication, Inclusivity, and Collaboration Enhancement were also reported as comprising an opportunity. ChatGPT was said to enhance inclusivity through multilingual support, assistance for people with learning disabilities, and cost-effective education access. Academics noted that ChatGPT could help nonnative English speakers refine papers to meet academic standards, thereby offering an alternative to paid services. Marketing professionals highlighted ChatGPT's role in global communication and customer support. In software development, it was seen as improving interdisciplinary collaboration by simplifying technical language. In addition, ChatGPT was noted for expanding educational accessibility as an affordable self-teaching tool, in particular for coding novices.

DISCUSSION

This research was motivated by the dearth of holistic studies that have investigated the potential uses of general-purpose generative AI tools, such as ChatGPT, in real-world contexts. Although previous research has highlighted the system's transformative potential, its early adoption was largely unsupervised, with users and managers receiving minimal guidance in navigating this highly malleable information technology. Our findings shed light on the dominant tasks, perceived opportunities, and challenges of malleable AI in knowledge work.

We have identified three dominant cross-professional ChatGPT tasks: content and copy creation, research and data analysis, and idea generation. Although certain tasks demonstrate a clear domain-specific use of AI, such as coding in software-related professions, other tasks, such as idea generation and brainstorming, have been widely adopted across disciplines. Whereas the former tasks closely align with the core activities of specific professions, demonstrating a high degree of specialization in AI usage, the latter tasks are largely universal and relevant across a wide range of professional contexts, highlighting the cross-disciplinary adoption of AI tools. This duality emphasizes AI's capacity to address both field-specific needs with tailored applications and broader tasks that transcend occupational boundaries.

Theoretically, the malleability of LLMs challenges traditional notions of task–technology fit, which typically emphasize the alignment between the characteristics of a technology and the requirements of a specific task to achieve optimal performance. Classical task–technology fit models often assume that a particular technology is best suited to a narrow range of tasks, suggesting a relatively fixed match between tool and function (Goodhue & Thompson, 1995; Strong et al., 2006). However, our findings illustrate that as a highly malleable, general-purpose AI, ChatGPT defies this assumption by enabling a wide range of applications that span both context-specific and cross-functional purposes. Its usefulness is not inherently tied to fixed technical features but is instead dynamically shaped by user intent, prompt design, and domain expertise. This indicates a shift from a technology-centric to a user- and interaction-centric view of task–technology alignment, where the effectiveness of AI depends less on static properties and more on how humans adapt and repurpose it within their professional contexts.

This malleability also carries significant practical implications for the implementation of LLMs. Unlike many organizational software tools associated with a set of a priori defined benefits (e.g.,

enterprise resource planning tools), the specific benefits of malleable software such as ChatGPT materialize only once it has found its place in daily work practices (Richter & Riemer, 2013). Hence, the implementation of such software should not rely on predefined measurable benefits determined in top-down fashion; instead, research suggests the adoption of a dynamic, bottom-up approach that supports experimentation and learning, allowing users to uncover and shape context-specific benefits. To support this process, researchers recommend using common tasks as a starting point to develop more fine-grained use cases for a specific organization in this sector (Richter & Riemer, 2013). Against this backdrop, our findings on the types of tasks professionals already use AI for can help organizations implement ChatGPT and similar tools by identifying and describing tasks that can serve as a starting point for further contextualization.

By understanding the specific use cases across various professional fields, organizations can effectively plan, conceptualize, and implement measures that are tailored to the unique needs of each domain. For example, tailored educational programs focused on practical applications, such as developing prompting guidelines, designing human–AI collaboration modes, or selecting appropriate AI tools, can enhance employees' ability to effectively use AI.

An illustrative example of the malleability of ChatGPT is AI-powered code generation. Programmers were among the first to experiment with the generative capacities of LLMs, quickly discovering their aptitude for code generation, an ability rooted in the vast amount of publicly available code included in their training data (Chen et al., 2021). This use case exemplifies the malleability of tools like ChatGPT, which has since evolved into specialized models, such as OpenAI's Codex, and practical tools, such as GitHub Copilot. Among digital companies, for instance, AI-generated coding—so-called "vibe coding"—has become a prevalent phenomenon, with 25% of current start-ups within Y Combinator's leading start-up accelerator program having generated 95% of their code bases using generative AI (Mehta, 2025). What began as experimentation by programmers soon gave rise to an entirely new work paradigm in code development, which also had imminent consequences for the skill sets investors or employers seek when building or evaluating start-up teams. This paradigm shift is likely to affect established companies soon as well; at present, up to 30% of the code of major technology companies (e.g., Microsoft) has been written by AI (Zeff, 2025). At first, this could be considered just another efficiency measure, but the speed advantage will quickly accumulate in the fast-iterating product development cycles, and eventually the compounding effect will result in a significant competitive advantage (McGrath, 2024).

With respect to the challenges inherent to the use of AI, our findings highlight several challenges professionals face when integrating AI into their work processes, which can be grouped into four key categories: personal (e.g., job replacement anxiety, deskilling), technical (e.g., accuracy, plagiarism), organizational (e.g., privacy concerns, implementation challenges), and macro level (e.g., AI biases, social inequality, economic disruption). Although certain challenges, such as accuracy and implementation, are common across domains, others are strongly shaped by the specific demands and norms of individual professions.

Among all of our respondents, accuracy and reliability concerns were the most frequently mentioned barriers. Professionals across fields reported a hesitancy to rely on AI because of the risk of inaccurate, biased, or unreliable outputs, which can undermine decision making and task quality. These concerns align with previous research that identified inaccuracy as a key obstacle to broader AI adoption (Abdullah et al., 2022; Hou & Ji, 2024). However, OpenAI (2024, 2025) has started to directly address these limitations by improving the accuracy of its models through newer features and architectures, indicating a positive trajectory. Challenges with implementation and adoption also correspond with earlier studies (e.g., Choi et al., 2023; Rahmaniar, 2024), in particular around adapting workflows, integrating AI into legacy systems, and managing data security risks. These challenges are especially pronounced among software professionals, our sample's most frequent users and earliest adopters of advanced LLMs, which may indicate that the barriers intensify with increased reliance on the tool. At the same time, the findings underscore that AI-related concerns are also highly domain

specific. In academia, for example, plagiarism and questions of originality are particularly salient. This reflects the field's emphasis on intellectual rigor, novel contributions, and empirical soundness. In contrast, the primary concern in software development is functional efficacy: Efficiency, problem solving, and deliverables matter more than theoretical originality. Although plagiarism is a major ethical concern in academic settings, raising alarm among educators, journal editors, and researchers about AI's role in outsourcing key scholarly tasks, it is perceived as relatively unproblematic in the software sector.

Our study contributes to ongoing theoretical debates (Dwivedi et al., 2023; Kasneci et al., 2023) by capturing practitioners' perspectives on these challenges. These insights can help leaders in different domains focus on the barriers most relevant to their users and develop clear, tailored strategies for addressing them. For example, in academic contexts transparent guidelines on the ethical use of AI may help reduce ambiguity and foster responsible adoption. In software professions, more immediate concerns, such as job security and effective implementation strategies, are likely to become even more pressing in the near future (Kaufman, 2025).

By identifying opportunities, we contribute to a better understanding of ChatGPT usage motivations by highlighting its professional benefits. We found four key areas of opportunity: efficiency-related benefits, such as time and cost savings; quality improvements related to enhanced content and creative output; self-improvement and continuous learning opportunities, reflecting ChatGPT's role in skill development and personal growth; and collaboration benefits, such as enabling global communication and fostering inclusivity. Efficiency gains (71%) and improved content quality (40%) were the most frequently cited benefits of ChatGPT. These results echo prior findings indicating that LLMs can enhance both speed and output quality across tasks such as coding (Fan et al., 2023; Hou & Ji, 2024; Huang et al., 2024), creative processes (Boussioux et al., 2024; Joosten et al., 2024; Polster et al., 2024), and strategic consulting (Dell'Acqua et al., 2023).

The benefits highlighted by professionals across different fields reflect the distinct characteristics of their job profiles. For example, although efficiency gains were the most prominent benefit across all domains, they were particularly emphasized by software professionals. This focus may be attributed to the transparent, time-sensitive nature of agile software development practices, which emphasize streamlined workflows, effective prioritization, meticulous planning, and time-saving processes (Abrahamsson et al., 2017). In contrast, academic professionals were more likely to highlight learning enhancement as a key benefit, reflecting their focus on continuous knowledge acquisition and intellectual growth. Meanwhile, cost efficiency is a significant driver in marketing professions, likely because of the field's emphasis on budget optimization and maximizing return on investment.

The efficiency and quality gains achieved through the use of AI have several practical implications. First, employees may consider AI a secret weapon of sorts to set themselves apart from the internal competition. Thus, being an early adopter of the latest models and tools, while continuously acquiring and refining relevant skills, is becoming increasingly crucial. Second, the advantage AI holds could be used to improve work–life balance, for example, by reducing working time. Anecdotal observations indicated that employees use the efficiency boost to create some buffer time, which might eventually lead to a significant amount of additional organizational slack (Marlin & Geiger, 2015; Nohria & Gulati, 1997). For organizations, an important task will be to regulate the slack and optimize the positive effects while avoiding negative consequences. This could be achieved by reviewing and, where necessary, redesigning existing job descriptions to adapt the amount of slack from AI augmentation to what is most beneficial for the individual and the organization.

In an email to employees, Micha Kaufman (2025), CEO of Fiverr, a leading platform that connects companies and freelancers, described the shift in an AI-augmented working world: "You must understand that what was once considered 'easy tasks' will no longer exist; what was considered 'hard tasks' will be the new easy, and what was considered 'impossible tasks' will be the new hard." The statement highlights how companies may address AI-driven efficiency and quality gains and communicate their increased performance expectations to prevent excessive slack. In a similar vein, some

companies have stopped replacing employees lost through natural attrition or retirement, instead turning to AI systems to absorb and automate these roles (Bloomberg, 2024).

This study has limitations that should be considered. The first one is the focus on ChatGPT. Although this ensured higher rigor and increased the specificity of our findings by preventing responses from being diluted by replies related to other generative AI tools, it may limit the generalizability of the findings to other generative AI applications. Moreover, the rapid evolution of ChatGPT is a limitation. Our data were captured in September 2023, and some issues raised by respondents may now be either less relevant or even more relevant. For example, several participants noted the model's poor mathematical performance, but follow-up studies have shown steady improvements in these operations (Frieder et al., 2023; Wei, 2024; Zhong et al., 2024). Last, we focused on U.S. respondents for various reasons explained in the "Methodology" section. Although the selected sample provided valuable insights, future studies may expand this analysis to include other regions and increase the generalizability of the results.

CONCLUSION

This study advances the understanding of how professionals across sectors adopt and adapt malleable generative AI tools such as ChatGPT in daily work. In contrast to prior research that has focused on specific domains, tasks, general workforce, or theoretical angles, we have provided a cross-professional, empirical perspective that is based on qualitative survey data from 78 employees in AI-exposed fields. We have identified common use cases, such as content creation, research, and idea generation, as well as recurring benefits and challenges. By mapping these across domains, we have revealed both cross-cutting patterns and domain-specific variations in how ChatGPT is applied.

Our findings challenge traditional notions of task–technology fit by illustrating how ChatGPT's effectiveness is shaped less by fixed technical attributes and more by user intent, creativity, and contextual adaptation. As such, ChatGPT exemplifies a new class of general-purpose, user-configurable technology that demands revised frameworks that account for interpretative flexibility and sociotechnical co-evolution. In a practical sense, this study highlights the importance of bottom-up integration strategies rooted in observed work practices and common task structures, rather than top-down deployment based on predefined benefits. Although benefits such as efficiency gains and improved output quality were evident across professional sectors, concerns around overreliance, plagiarism, and job replacement varied by context and must be proactively addressed. We argue that organizations that embrace the malleability of LLMs and invest in task-aware integration, training, and governance will be better positioned to harness AI for sustainable value creation.

FUNDING STATEMENT

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

COMPETING INTERESTS

The authors of this publication declare there are no competing interests.

PROCESS DATES

This manuscript was initially received for consideration for the journal on 04/04/2025, revisions were received for the manuscript following the double-anonymized peer review on 05/25/2025, the

manuscript was formally accepted on 05/25/2025, and the manuscript was finalized for publication on 07/25/2025.

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ACKNOWLEDGMENT

The authors ensured that the study adhered to the ethical guidelines of the German Research Foundation (Deutsche Forschungsgemeinschaft). In accordance with these guidelines, formal ethics committee approval was not required because the study did not involve patients, individuals with special protection needs (e.g., by applying age-based prescreening), or procedures that posed significant psychological or physical risks. All respondents participated voluntarily, were informed of the study's purpose, and provided their consent to participate. All responses were collected anonymously.

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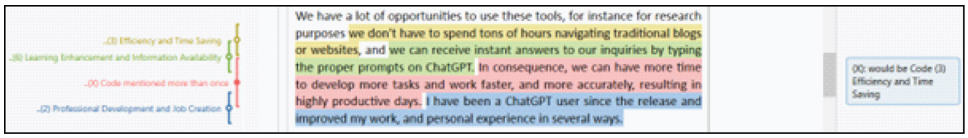
APPENDIX

Figure 1. Coding Snippet (Opportunities in the Marketing Sector)



Several distinct codes (e.g., Codes 2, 3, and 6) can be assigned to a single answer, but the same code cannot appear more than once in the same answer. If a segment would again fit under a previously used code, it was marked with an X instead of reassigning that code.

Table 7. Explanations and Examples of Tasks

Code	Explanation	Example
Content and Copy Creation	Generating, refining, and structuring text across various platforms and formats	"I mostly use it for tweaking marketing copy/social media copy, but I also use it for rough drafts and occasionally complete copy generation."—Marketing director, 46
Research and Data Analysis	ChatGPT assists various types of data analyses and different research stages.	"ChatGPT aids me in . . . fact-checking, citation assistance, efficient literature review, [and] peer review preparation."—Postdoctoral associate, 41
Idea Generation	Generating new ideas, brainstorming creative solutions, and providing initial inspiration for various tasks and projects	"ChatGPT is also helpful when lesson planning, particularly when I am trying to come up with possible advancing questions to ask the students after they complete an activity."—High school math teacher, 24
Software Development and Coding	Creating, optimizing, and debugging software and codes as well as more general software-related troubleshooting	"I mostly use it . . . to solve bugs I'm having issues with."—Junior Python developer, 32
Knowledge Acquisition	Acquiring knowledge and facilitating information retrieval	"[I use ChatGPT] to reformulate a paragraph in simple terms. Sometimes, academic papers can be hard to read and understand. ChatGPT helps me to 'dumb down' sentences."—Research associate, 32
Stakeholder Communication	Drafting, refining, and managing communications with stakeholders	"I use ChatGPT to help write out customer-facing communications regarding platform support for our product. I like to use ChatGPT to create the base of my draft messages. . . . Then I'll tweak them."—Customer experience professional, 38
Administrative and Operational Support	ChatGPT helping perform and manage administrative duties and operational coordination tasks	"I use it to . . . go through lists of applicants and sort them by requirements and their particular strengths/weaknesses."—Network support specialist, 40
SEO and Keyword Research	Improving SEO, visibility, and performance (e.g., with keyword suggestions)	"I use ChatGPT to help me with keyword research. ... ChatGPT also helps me . . . answer common search queries to improve our website's visibility on search engines."—Analyst, 40

Note. SEO = search engine optimization.

We have shortened some of the examples to provide a clearer overview. In most cases, participants further elaborated on tasks; for example, the following is the full elaboration of the 40-year-old analyst's example in Table A1:

> *I use ChatGPT to help me with keyword research. I input topics and get suggestions for popular keywords, which makes it easier to optimize website content. ChatGPT also helps me generate ideas for blog posts and answer common search queries to improve our website's visibility on search engines.*

Table 8. Explanations of Challenges

Code	Explanation
Accuracy and Reliability Issues	Factual inaccuracies, outdated information, and ChatGPT's tendency to generate convincing but false data ("hallucinations"; Maynez et al., 2020)
Operational and Implementation Challenges	Practical difficulties of effectively integrating and using ChatGPT within existing workflows
Overreliance and Deskilling	Dependence on ChatGPT that undermines the development of fundamental skills, problem solving, and critical thinking
Originality and Plagiarism Concerns	Risk of generating generic content, potential copyright infringement, and ethical dilemmas related to authorship and authenticity
Job Replacement Anxiety	Concerns that ChatGPT may automate tasks traditionally performed by humans, potentially leading to job displacement, salary reductions, or diminished value of certain roles within their sectors
Privacy and Data Security Concerns	Fear that data could be compromised or that compliance regulations will be violated
Social and Economic Impact	Broader consequences of integrating ChatGPT across various sectors, including effects on societal norms, economic structures, and professional landscapes. Excludes direct job-related consequences, which are addressed in the "Job Replacement Anxiety" category.
AI Bias and Misuse	Potential to exhibit prejudiced behaviors or to be used to propagate misinformation, manipulate opinions, or violate ethical standards

Note. AI = artificial intelligence.

Table 9. Explanations of Opportunities

Code	Explanation
Efficiency and Timesaving	Ability to streamline tasks, increase productivity, and reduce the time required for routine and various activities
Content Proficiency and Data Intelligence	Enhance content quality, accuracy, and relevance while enabling effective data analysis for informed decision making
Learning Enhancement and Information Availability	Ability to improve comprehension, provide accessible information, and facilitate the knowledge acquisition process
Innovative Thinking and Creativity	Ability to inspire new ideas, enhance brainstorming, and support creative problem solving across various sectors

continued on following page

Table 9. Continued

Code	Explanation
Professional Development and Job Creation	Enhance individual skill sets, improve job performance, and create new employment opportunities across various sectors
Cost Efficiency and Profit Maximization	Ability to streamline operations, reduce operating costs, and enhance revenue generation
Global Communication, Inclusivity, and Collaboration Enhancement	Leveraging ChatGPT to eliminate language barriers, provide equal access to information, and facilitate collaborative work in diverse and globalized environments

Pedro Schötteler Libera is a Master's student in Industrial Engineering with a Management focus at the Nuremberg Institute of Technology, having commenced his program in March 2024. He is particularly intrigued by how artificial intelligence is reshaping both corporate environments and society at large. As part of his academic work, Pedro is examining the integration of ChatGPT in the modern workplace, highlighting potential tasks, challenges, and opportunities. In addition to his studies, he collaborates with an AI team in a forward-thinking company, applying emerging technologies to real-world scenarios. Looking ahead, he plans to pursue a PhD to further explore the intersection of technology, management, and innovation.

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